# SUMMARY STATISTICS

new 1.7–1.8 • old 1.7–1.8

## ZONE A · THE FORMULAS

$$\bar{x} = \frac{\sum x_i}{n}$$

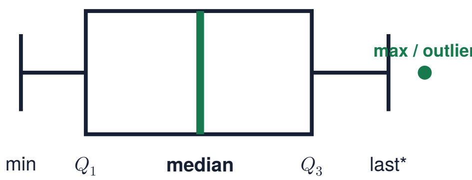
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$\text{IQR} = Q_3 - Q_1$$

outlier if $x > Q_3 + 1.5(\text{IQR})$
or $x < Q_1 - 1.5(\text{IQR})$

### A BOX PLOT IS A FIVE-NUMBER STORY




*The whisker stops at the last non-outlier. The maximum here is the separate point.

**CENTER + SPREAD WORK AS A PAIR**

symmetric, no outliers $\rightarrow$ $\bar{x}$ and $s$
skewed or outliers $\rightarrow$ median and IQR

## ZONE B · WHAT THE SYMBOLS MEAN

### SAMPLE

$\bar{x}$ = sample mean $s$ = sample SD
five-number summary: min, $Q_1$, median, $Q_3$, max

### POPULATION + RESISTANCE

$\mu$ = population mean $\sigma$ = population SD
**resistant:** median, IQR **not resistant:** mean, SD, range

## ZONE C · SAY IT IN WORDS

*"Typically, ______________ varies from the mean by about _______ _______."*

variable in context value of $s$ units

## ZONE D · WATCH OUT

### THE DENOMINATOR

For a **sample** variance or SD, divide by $n-1$, not $n$.

### TRANSFORMATIONS

Adding a constant shifts the center but not the spread. Multiplying by a constant scales **both**.

see WS old u1_l7–u1_l8